1.0 Overview

1.1 Table of Contents

1.0 Overview
- 1.1 Table of Contents
- 1.2 Purpose
- 1.3 Scope
- 1.4 Responsibilities

2.0 Definitions
3.0 Requirements
- 3.1 System Life Cycle Methodology

4.0 SLC Phases
- 4.1 Planning Phase
- 4.2 Requirements Analysis Phase
- 4.3 Design Phase
- 4.4 Construction and Coding Phase
- 4.5 Testing and Implementation Phase
- 4.6 Operations and Maintenance Phase
- 4.7 Decommissioning Phase

5.0 Appendices

1.2 Purpose

The purpose of this document is to provide the computer System Life Cycle (SLC) process requirements for non-product software deployed and / or maintained by Business and Technology Services (BTS). Use of this process is intended to produce applications that are developed and certified/validated for their intended use.

Implementation Strategy: IT Projects with an approved application change request prior to the effective date of this BTSQC09.05 version may complete execution of the project utilizing the BTSQC09.05 version effective at the time of change request approval.

1.3 Scope

In-Scope

This document applies to the development, maintenance and procurement of all non-product software applications developed and / or managed by BTS IT, including:

- Developed or purchased software applications
- Externally hosted solutions (Software-as-a-Service / subscription)
- Mobile Applications (non-medical) made available to consumers via digital devices (e.g., smartphones)
- Automated Process Control systems (APCs)

Completion of SLC deliverables for Non-Regulated applications is encouraged for good software development/deployment practices. The following deliverables are required for Non-Regulated applications:

- System Risk Classification (SRC)
- Business Impact Analysis (BIA), as applicable
- Data Lifecycle Management (DLM) Assessment, as applicable
- Data Protection Plan, as applicable

This information is confidential to Abbott. The user is responsible for using the appropriate version of this document.

Abbott sites serving markets with more rigorous regulations than this process must use those requirements in addition to this process.

Out of Scope

The following are out of scope:

- Product software/applications
- Qualification and verification of IT Infrastructure and hardware following QC10.11 - Infrastructure Qualification Process

Additionally, the following software development tools are excluded from this process:

- Word processing software (e.g., Microsoft Word) when the output is reviewed, approved, and controlled using record control procedures
- Computer Aided Design (CAD) applications when the output is verified or validated
- Third-party software components that only function when compiled into custom code where the created code output is verified or validated (e.g., spell checking components, Crystal Reports, Windows forms managers)
- Source code editors, compilers, linkers, assemblers, and interpreters when the code generated by these systems is validated / certified for its intended use
- Middleware / layered software which is indirectly validated / certified with the application
- Network management systems, operating systems, and operating system environments when they are used during validation
- Computer systems that are integral to Facilities, Utilities, or Equipment. (This software is validated with the associated Facility, Utility, or Equipment as a system.) and
- Output (e.g., report, spreadsheet, statistical calculation) that is verified in place of validation.

***Note**:* Verification requires review of inputs, formulas, outputs, and appropriate verification / approval for the project, based on the methodology selected.

1.4 Responsibilities

The table below lists the functions for a cross-functional non-product validation lifecycle team.
All team members involved in the system life cycle process are responsible for compliance with this process and ensure that the application is compliant with applicable Abbott policies, processes, procedures, and global regulations.

The attachments to this process, Appendix A: Author and Approval Matrix for Quality Applications and Appendix B: Author and Approval Matrix for Regulated (Non-Quality) are to be used to determine deliverables, suggested authors, and approvers.

Roles within this SLC reflect functional roles rather than specific job titles. The roles below are to be fulfilled by Abbott (including staff augmentation) or by delegate. Training for SLC practitioners is based on the functional roles identified in this document.

This information is confidential to Abbott. The user is responsible for using the appropriate version of this document.

Function/Role	Responsibility
Process Owner	
Business Process Owner (BPO) Note: Business Process Owner and System Owner can be performed by the same individual	• Has ultimate responsibility for ensuring the computer system and its operation is compliant and fit for intended use in accordance with applicable SOPs • Responsible for or knowledgeable of the business process and answers auditor questions on use of the system • Defines the scope and translates the business processes into user requirements • Defines intended use and any performance requirements • Owns and assumes business risk for the computer application • Owns the application data and responsible for the content accuracy • Coordinates input from other groups (e.g., regulatory affairs, information technology, safety, legal, etc.) • Ensures compliance to Abbott and regulatory requirements and Corporate Brand Standards • Provides adequate resources (including SMEs and personnel) to support development and operation of the system • Executes User Acceptance Testing (UAT) and may execute Operational Qualification (OQ) testing • Ensures that Standard Operating Procedures (SOPs) required for operation of the system exist, are followed, and are periodically reviewed • Ensures adequate training for business end users • Accepts the system at handover (release) • Identifies opportunities for process improvement • Ensures changes are controlled, approved, and managed • Reviews assessment / audit reports, responding to findings, and takes appropriate actions to ensure ongoing compliance • Determines the record retention requirements in accordance with the Global Records and Information Policy (I1-02) and Data Lifecycle Management (DLM) Policy (I1-04)) • Determines business impact of an outage per Business Impact Analysis (BIA) ***Note:*** On larger or complex applications, there may be multiple Business Process Owners. Business Relationship Managers (BRM) can represent one or more Business Process Owners.

This information is confidential to Abbott. The user is responsible for using the appropriate version of this document.

Compliance Function	
SQA Lead (i.e. SQA Manager)	• BTS software quality assurance manager that reviews and approves key project deliverables for quality applications (SQA activities) as defined in the Validation Plan, or equivalent • Assures the system is compliant with applicable policies, processes, procedures, and global quality regulations • Provides subject matter expertise on GxP regulations for quality and final authority of application and functional GxP impact • Responsible for providing guidance to BPO regarding defending the validation package in an audit • Final authority on quality decisions and approval / release of a GxP computer system • May fulfill the role of the SQA and Compliance Analyst (CA) • Assigned by management, responsible for assuring compliance to GxP requirements for software quality
Software Quality Assurance (SQA)	• A quality representative that reviews and approves key project deliverables for quality applications (SQA activities) as defined in the project Validation Plan or equivalent • Assures that the system is compliant with applicable Abbott policies, processes, procedures, and global quality regulations • Escalates compliance concerns to the SQA Lead • May fulfill the role of the Compliance Analyst (CA)
Compliance Analyst (CA)	• A compliance representative that reviews and approves project deliverables for Regulated (Non-Quality) Applications as defined in the Verification Plan or equivalent • Assures that the system is compliant with applicable Abbott policies, processes, procedures, and global regulations • Escalates compliance concerns to the SQA Lead
Subject Matter Expert (SME) (i.e., Regulatory SME)	• Assures that required deliverables are compliant with appropriate regulations and adequate from the perspective of their individual area of expertise • SMEs may be compliance experts in areas such as Sarbanes Oxley Express (SOX), Quality, Privacy or other (i.e., Technical or Mobile SME)

This information is confidential to Abbott. The user is responsible for using the appropriate version of this document.

System Owner	
IT Support Owner ***Note:*** IT Support Owner and Primary IT Lead roles fulfill the System Owner role	• Provides adequate resources (including SMEs and personnel) to support development and operation of the system • Responsible for owning the system technology, including hardware, applications, software code, interfaces, operating systems, etc. • Responsible for the application support including availability, maintenance, recertification, back-up and restore, disaster recovery, integrity and security of data residing on the system • Responsible for providing production support of applications in compliance with governing procedures • Responsible for reviewing production change control, transition to support and deployment deliverables • Responsible for providing guidance to BPO regarding defending the validation package in an audit • Responsible to administer the process for retention, transfer, and disposal of records per the Abbott Records Retention Schedule • Responsible for Production Incident, Problem and Defect Management • Responsible for maintaining the system in a certified state • Ensures applications are logged and current in application inventory • Ensures application adheres to Abbott's Security Requirements
Primary IT Lead (i.e., Delivery Owner) ***Note:*** IT Support Owner and Primary IT Lead roles fulfill the System Owner role)	• Accountable for design and project delivery of applications to meet the intended use of the business as defined in the requirements • Ensures alignment with Corporate Brand Standards • Assures that appropriate expertise is available to achieve system certification / validation • Responsible for transition to support with the IT Support Owner. • Responsible for reviewing and approving project deliverables for system certification / validation • Responsible for testing and test defect management in projects. Technical responsible for the system change, design, and validation strategy • Accountable for technical expertise of the Requirements, Issues and Defects; responsible to engage technical resources as necessary • Responsible to administer the process for retention, transfer, and disposal of records per the Abbott Records Retention Schedule • Ensures application adheres to Abbott's Security Requirements (as applicable)

This information is confidential to Abbott. The user is responsible for using the appropriate version of this document.

Validation Lead (VL) ***Note:*** The Val Lead role may be filled by any member of the IT Delivery Team with the appropriate knowledge and experience	• Works under the IT Delivery Owner • Bridges between the IT Delivery team and SQA, facilitating project decisions under guidance of SQA in alignment with the Validation Plan and risk-based test assurance strategy • Role required for Quality applications and must be knowledgeable in Software Lifecycle and software validation principles • Assists with the development of system requirements and specifications to ensure requirements are specific, accurate and testable • Assists project teams with and / or develops and maintains validation deliverables, including plans, test scripts and user acceptance tests • Manages all testing activities for the project, including the execution of test plans and coordination of automated testing and routes software lifecycle records for review / approval • Facilitates test defect investigation, retest, and resolution of defects • Guides implementation teams in the proper execution of validation package requirements and ensures documentation is completed correctly and acts as the “gatekeeper” to verify documents are approved before related validation activities commence • Escalates compliance concerns / decisions to appropriate project team members and / or regulatory SMEs and SQA Lead
Project Manager	
Project Manager	• Coordinates day-to-day project activities • Coordinates project team members to ensure the applications are designed to meet the intended business use
Technical Representative	
Architect / Technical (Technical Manager)	• Ensures controls are implemented to support the Data Protection Plan as appropriate • Ensures that development, support, and maintenance practices are appropriate and compliant for applications • Ensures the impact of the change is accurately evaluated for applications. • Technical responsibility providing input for the system change, design, and validation / testing strategy • Responsible for ensuring the design is appropriate to achieve the defined requirements (URS/FRS) • Ensures Source Code Review is performed as applicable • Identifies functions that necessitate Unit Test cases and ensures Unit Test cases are executed, as applicable • May fulfill multiple roles for IT Technical Team

This information is confidential to Abbott. The user is responsible for using the appropriate version of this document.

IT Technical Team	• Defines and documents, as applicable, system architecture, configuration, and design aspects including components / functions, databases, interfaces, cybersecurity, security and relevant system information for development and maintenance • Updates documentation as needed as part of each major release / upgrade • May act as an IT Peer Source Code Reviewer if not the Code Author and understand the software source code and underlying business process sufficiently to provide a meaningful review • Identifies functions that necessitate Unit Test cases and executes Unit Tests
Review Board	
Validation Review Board (VRB) or equivalent	Representatives: System Owner / Business Process Owner, SQA Lead and Technical. An ad-hoc review board may be created to resolve project issues asneeded to: • Assess applications for regulatory status and certification requirements • Identify subject matter experts needed for regulated applications • Review and approve deliverables for system certification ***Note:*** VRB disputes should be elevated to the immediate manager of each dissenting VRB representative. Global projects may require VRB representation from multiple sites.

2.0 Definitions

Click this link to access the Abbott Glossary.

Note: The following terms are document specific and are included for reference in this document only.

Term	Definition
Complexity	The implementation strategy for a given function or system (Commercial-off-the-Shelf (COTS), Configured, and/or Custom)

This information is confidential to Abbott. The user is responsible for using the appropriate version of this document.

Term	Definition
Computer System Lifecycle	The computer system lifecycle encompasses all activities from initial concept to retirement. The lifecycle for any system consists of seven (7) major phases: • Planning • Requirements Analysis • Design • Construction and Coding • Testing and Implementation • Operations and Maintenance • Retirement(Decommissioning)
Computer System Validation	The process of documenting that a computer system meets a set of pre-defined system requirements; executed to ensure accuracy, reliability, performance, consistency, and the ability to discern invalid or altered records, which is a critical requirement of electronic record compliance
Data Integrity	Maintenance and assurance of the accuracy and consistency of data over its entire lifecycle, from creation through destruction, and is a critical aspect to the design, implementation and usage of any system which stores, processes, or retrieves regulated data
Decommissioning	Permanent removal of a system or component from its operational environment, including removal of support from an operational system or component, removal from operations
Hypercare	The period of time immediately following a Go-Live, when the IT Delivery Team is retained and provides an elevated level of support to ensure seamless adoption of a new or upgraded system. Change management procedures apply
Qualification / Testing	Documented formal and systematic evidence that applications are suitable for intended use, based on design criteria (process requirements or equivalent) and the verification work that is performed throughout the delivery process, including testing performed
Requirement	Need or expectation that is stated, generally implied, or obligatory; translated into requirements specifications
Risk Assessment	A systematic process of assessing the functional risks of the system measured against patient safety, product quality, data integrity and regulations with an intent to drive the type and level of testing for assurance of proper functionality
Scenario Based Testing	Dynamic testing in which the tester's actions are not prescribed by detailed, step-level instructions in a test case. High level test objectives for features and functions are established

This information is confidential to Abbott. The user is responsible for using the appropriate version of this document.

Term	Definition
Scripted Testing	Defined testing in which the actions are prescribed by written instructions (often with detailed step or click action items) in a pre-approved test script
Specification	A document that specifies, in a complete, precise, and verifiable manner, the requirements, design, behavior, and / or other characteristics of a system or component, and often, the processes for determining whether these provisions have been satisfied
Test Assurance Level	A risk measure of a requirement's potential to impact patient safety, product quality, data integrity and regulatory impact measured against the requirement's implementation complexity; corresponds to the defined testing approach

3.0 Requirements

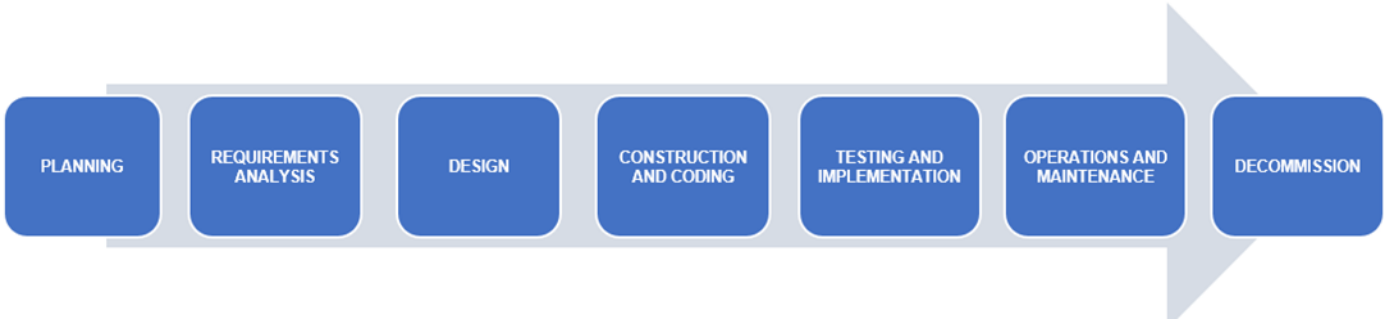
3.1 System Life Cycle Methodology

3.1.1 Process Description

A System Life Cycle (SLC) is used for computerized system validation / certification, to build in quality during the process, from initial idea until the decommissioning of the system. Validation is a form of certification for computer applications that impact Abbott quality.

3.1.2 SLC Phases

The methodology utilized must meet the minimum requirements established in the SLC. Established methodologies provide the framework for successful system development and maintenance (i.e., Waterfall, Agile, ASAP, etc.). The SLC may consist of the following common phases:



3.1.3 Templates

Templates drive consistency, compliance, and provide valuable instructions and content. Templates can be tailored but must meet the minimum requirements of the SLC Process and Procedures.
BTS SLC templates utilize:

- Red italicized text for instructions,
- Blue text as examples or content placeholders and
- Black text for content which must be addressed or justified if not applicable.

This information is confidential to Abbott. The user is responsible for using the appropriate version of this document.

BTS templates, created to support the SLC documentation process under this document, are available and should be used. Use of other templates may be addressed in tailoring details, including the use of Divisional templates, vendor documents or application specific documentation.

Note: Templates are listed with the corresponding deliverable in Appendix A and B with the suggested author.

3.1.4 Tailoring

Project teams may tailor deliverables based on the system risk classification, size and complexity of the project when justification is provided and approved. Justification must be provided for tailoring which removes a Required (R) or Required If Applicable (RIA) deliverable.
Use Appendix A or B to determine required project deliverables and minimum approvers.
Examples of tailoring include:

- Tailor and/or combine deliverables
- Separate deliverables (e.g. separate URS and FRS instead of combined)
- Combine or overlap phases
- Combine roles (Quality and Compliance roles cannot be combined with roles from another function)

Consult the Validation Review Board for guidance on tailoring deliverables.

3.1.5 Agile Framework

The Agile framework is an approach for requirements and solutions to evolve through the collaborative effort of self-organizing cross-functional teams. It promotes adaptive planning, evolutionary development, early delivery and continuous improvement. It encourages rapid and flexible responses to change. The Validation Plan / Change Request (CR) or equivalent should include details of Agile approach.

3.1.6 Software as a Service (SaaS) Applications

SaaS applications provide Abbott with the capability to use the supplier's application(s) running on a supplier's infrastructure and accessible from various client devices through a thin client interface such as a Web browser (e.g., web-based email). Abbott does not manage or control the underlying infrastructure, network, servers, operating systems, storage, or even individual application capabilities, except for limited user-specific application configuration settings.
Abbott is responsible for defining the intended use of its applications and the documentation that supports intended use, whether developed by a SaaS provider or Abbott resources. Project teams can include and use vendor documentation as part of the validation package. For applications classified as quality, responsibilities of the vendor must be defined within the project documentation, Service Level Agreement (SLA), contract, or equivalent.

3.1.7 Middleware or Infrastructure Software Changes Impacting Applications

Middleware or Infrastructure changes that have the potential to impact application functionality require regression testing of the application functionality at risk. Testing can be satisfied through functional testing, provided the functional test case exercises the middleware or infrastructure change. A collection set approach using selected applications may be applied. The testing strategy must be documented in the project's associated planning documentation (i.e., Validation/ Verification Plan). The applications within the collection set should take into consideration the functionality that may be essential to their business, compliance, or operations. Where regulated data is passing through middleware software, the target or source system's project documentation must include the appropriate end-to-end testing.

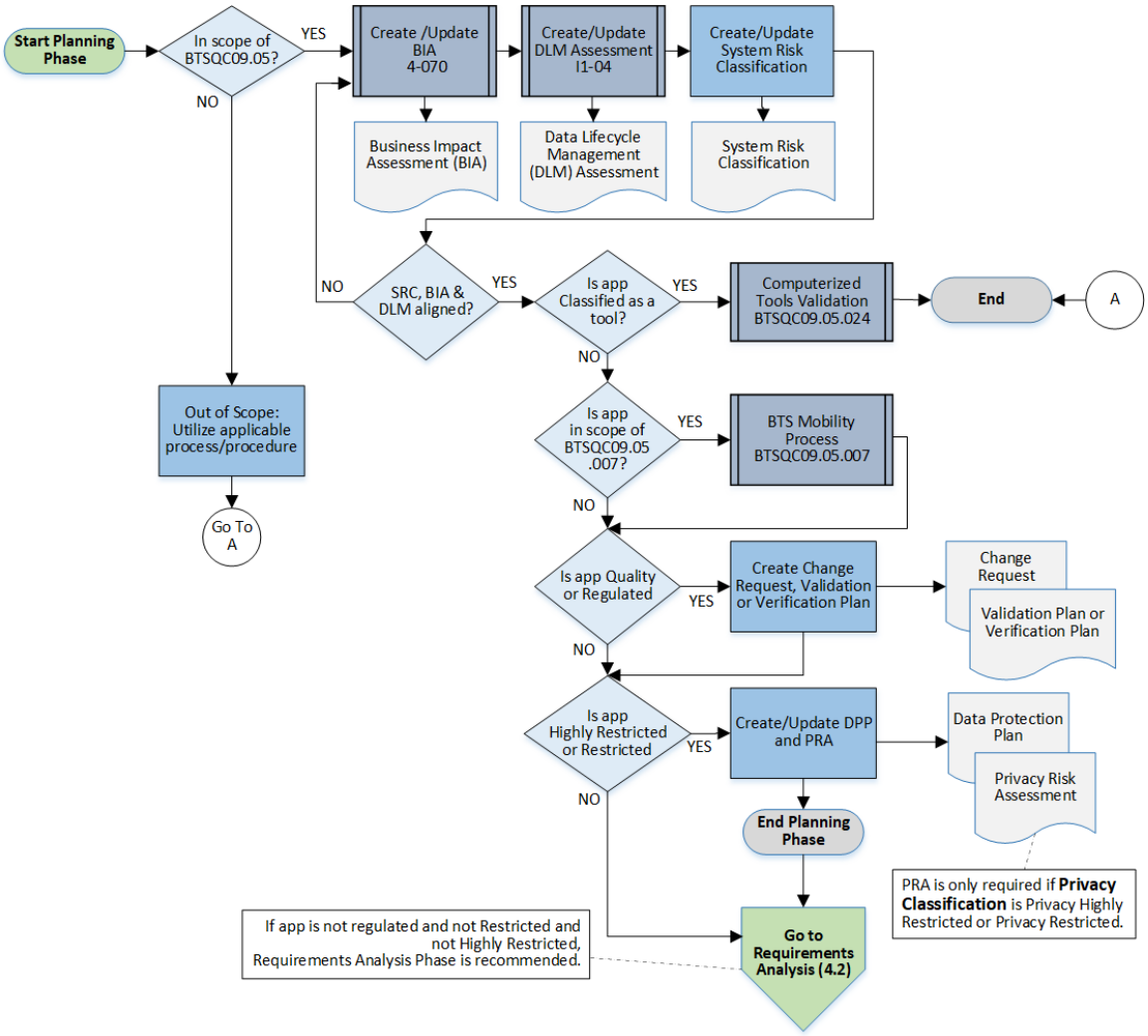
This information is confidential to Abbott. The user is responsible for using the appropriate version of this document.

4.0 SLC Phases

4.1 Planning Phase

This phase determines relevant activities and documentation based on intended use, size, scope and system complexity.

4.1.1 Planning Process Flow



This information is confidential to Abbott. The user is responsible for using the appropriate version of this document.

4.1.2 Deliverables

4.1.2.1 System Risk Classification (SRC)

Every application must be assessed for regulatory impact and validation needs. The System Risk Classification is the initial deliverable and must be completed for all applications. Once complete, the SRC is associated with the application throughout its lifecycle. While not routinely revised, the SRC must be assessed and maintained for changes in intended use, including regulatory applicability, complexity, data sensitivity, supporting vendors and hosting. There must only be one approved active SRC for a system. If a system validation package is migrated to a new repository, the SRC, or equivalent, in the source repository must reflect that it is no longer current or point to the target location in the new repository.

Upon approval of the System Risk Classification:

- Regulated applications continue under this process

Spreadsheets & Statistical Applications follow AQ090501.A, Spreadsheet and Statistical Software Application Verification and Validation, or equivalent. Refer to the purpose / scope and exceptions within AQ090501.A for applicability prior to creating the SRC.

Note: System Owner ensures the application is logged and current in application inventory.

4.1.2.2 Vendor Assessment Process

BTSQC11.01 -Technology Vendor Risk Management Process
BTSQC11.02 - BTS Supplier Quality Assurance Process
BTSQC19.21- Cybersecurity Vendor Risk Management Process

The vendor assessment process is a vital input into the SLC process. The process provides input into project planning. Further, a fully assessed software vendor may reduce the documentation and testing burden, particularly on SaaS applications and / or Commercial, Off-The-Shelf (COTS) requirements. Project teams can include and use vendor documentation as part of the validation package.

The vendor assessment process is divided by areas of focus; BTS Supplier Quality Management and Cybersecurity Vendor Risk Management. Both are performed to assess vendors' capabilities in meeting Abbott requirements and global regulations.

Note: The testing strategy must consider any gaps identified in the vendor assessment and treat risk through additional deliverables, testing or subjecting impacted requirements to a higher Test Assurance level.

4.1.2.3 Supplemental Documents

The following documents are owned and maintained by the Business Process Owner or equivalent.

a. Data Lifecycle Management (DLM)
Policy I1-04
A Data Lifecycle Management (DLM) Assessment is required for applications (per I1-04) and is executed to identify the necessary controls to ensure the computer system's records are

This information is confidential to Abbott. The user is responsible for using the appropriate version of this document.

accessible, readable, and retrievable throughout their records retention period, role-based access is applied, record modifications are restricted. The DLM and SRC must be aligned.

b. <u>Business Impact Analysis (BIA)</u>

<u>Policy 4-070</u>

BIA must be completed for every in-scope application to assess the potential qualitative (non-financial) and quantitative (financial) business impacts of the application being unavailable due to a significant disruption and / or disaster incident. The BIA must be reviewed and updated, as appropriate, in the event of significant changes to the application that require a change to the Recovery Time Objective (RTO) and / or Recovery Point Objective (RPO).

4.1.2.4 Disaster Recovery Plan (DRP)

<u>Policy 4-070</u>

Disaster Recovery Plan documents the disaster recovery solution to mitigate business impact of an unplanned outage as identified in the BIA. DRP is required when the BIA determines an RTO <=24 hours and must be reviewed and evaluated for updates when significant changes occur that impact application recovery.

4.1.2.5 Computerized Tool Validation

<u>Procedure BTSQC09.05.024</u>

Applications that are categorized as a computerized tool in the SRC may use BTSQC09.05.024 as a standard streamlined approach to qualify the computerized tool is fit for purpose.

4.1.2.6 Mobile Applications and Websites

<u>Digital Marketing Policy MKT03 and BTS Mobility Process BTSQC09.05.007</u>

Note: Consideration must be given to other Abbott Policy, Process or Procedure that impacts mobile applications and / or websites with quality and / or regulatory impact.

When developing and / or maintaining Mobile Applications and / or Websites, the following requirements must be met:

- Websites that pull approved content from another source must have the source, data and mapping defined and testing performed.
- Regulated websites shall be designed, developed, validated, implemented, maintained, and retired in compliance with all appropriate regulations including those for electronic records and electronic signatures. Risks to data integrity, content integrity, PCI, and privacy must be identified and managed by suitable controls.

If a mobile application or website shares data with a medical device, there are additional requirements that may need to be addressed. Follow the applicable product software SLC.

4.1.2.7 Change Request

A Change Request is required to evaluate impact of changes on an application. The change request should include the description/reason for change, impact of change, deliverables, testing strategy, and acceptance criteria. The impact of change should include but not limited to review of interfaces, architecture, infrastructure, documentation, security, SRC, DLM, BIA, application inventory, quality risks and regulatory risks.

This information is confidential to Abbott. The user is responsible for using the appropriate version of this document.

The Change Request establishes the Validation Documentation Package(s) in the associated electronic Validation Management System(s). This package creates a required reference that allows for organization of deliverables and all documents to be routed for approval and traced to a project. When decommissioning an application, a change request is used to complete all activities to remove the application from production and to close any project documentation.

4.1.2.8 Validation Plan or Verification Plan

A Validation Plan or Verification Plan is used to document the deliverables and activities needed to deliver a new application or for significant changes to existing applications (e.g., major enhancements due to extensive modifications to the underlying business process). The Validation Plan / Verification Plan may define a hyper-care period to ensure project teams are not disbanded so that issues that arise in the operational state can be rapidly triaged and addressed after initial Go-Live.

4.1.2.9 Data Protection Plan (DPP)

Policy I1-05

A Data Protection Plan (DPP) is intended to ensure compliance with local, state, and national laws, as well as regulations and contractual agreements regarding the use and protection of Personal Information. The DPP may require modification when a change includes privacy impact (i.e., a new use of Personal Information, change in geographic location of data) or business data sensitivity that is not documented in the existing DPP. The DPP must be approved before the project impacts any personal information (typically prior to testing or migration activities). The Data Protection Plan (DPP) supports the Secure Lifecycle Process (SLP) requirements to maintain the data securely. The DPP must be in alignment with the SRC/DLM. The DPP controls are confirmed via the Technical Control Test template.

***Note**:* A DPP must be created for any application with data privacy classification of Privacy Restricted or Privacy Highly Restricted or a business sensitivity classification of Highly Restricted or Restricted.

4.1.2.10 Privacy Risk Assessment (PRA)

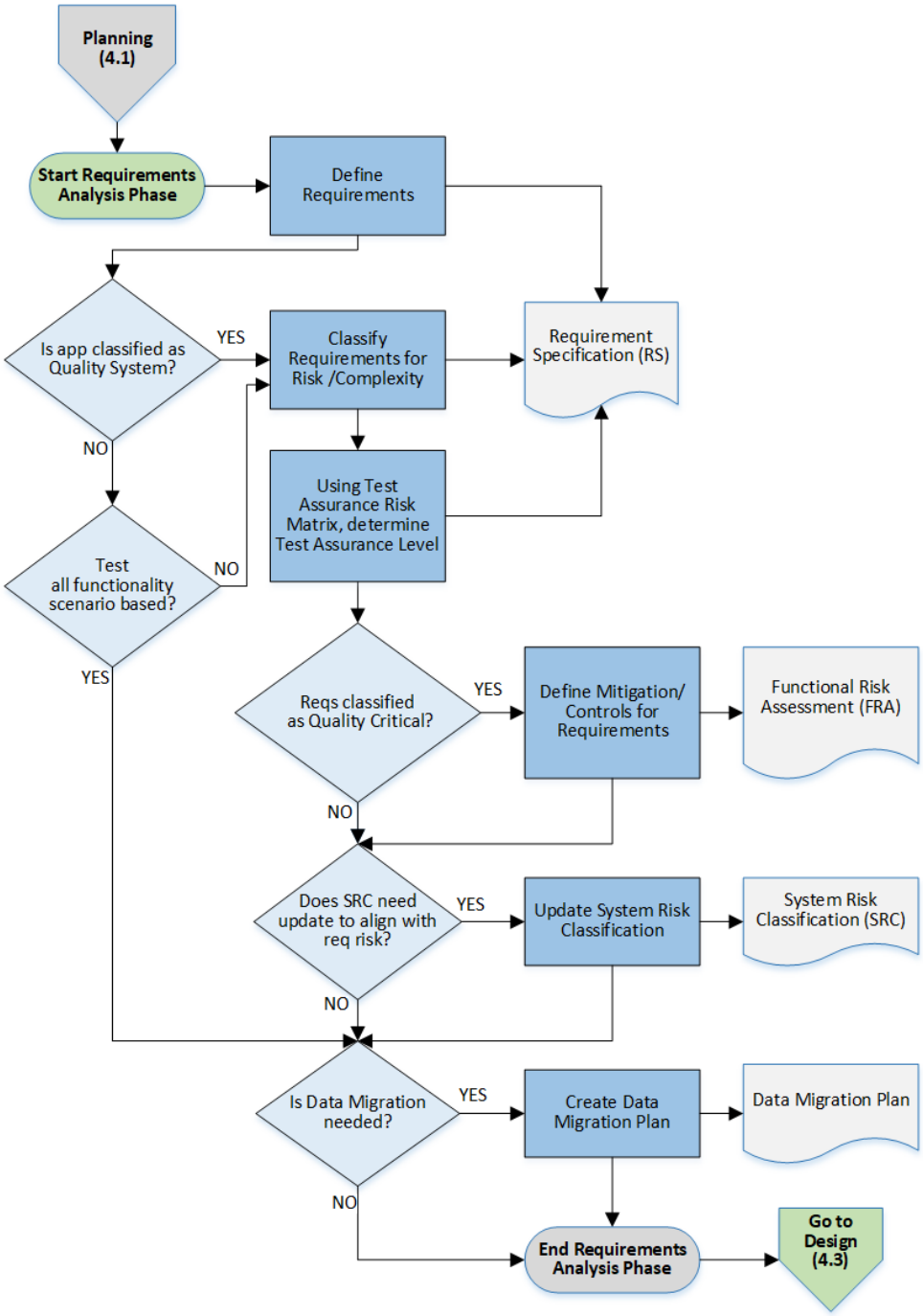
Process GLB-PRIVACY-RA-PROC

A Privacy Risk Assessment (PRA) is completed in the early stages of development to identify and mitigate privacy risks. The PRA incorporates the Privacy by Design controls to assess compliance with policy and is required for applications classified with **privacy classification** of “Privacy Restricted” or “Privacy Highly Restricted” per GLB-PRIVACY-RA-PROC. Services must also undergo a PRA if significant changes occur to the service.

This information is confidential to Abbott. The user is responsible for using the appropriate version of this document.

4.2 Requirements Analysis Phase

4.2.1 Requirements Process Flow



This information is confidential to Abbott. The user is responsible for using the appropriate version of this document.

4.2.2 Deliverables

4.2.2.1 Requirements Specification (RS)

The Requirement Specification (RS) defines the user and/or functional requirements and associated regulatory requirements for the application. The regulatory classification must align with the System Risk Classification (SRC).

Note: The Requirement Classification and Complexity are located in the Requirement Specification.

Note: A separate User Requirement Specification (URS) and Functional Requirement Specification (FRS) can be delivered as a tailored solution.

a. User Requirements (UR)

User Requirements (UR) define what the business requires the system to do from both functional and non-functional perspectives. User requirements capture the intended use. User requirements are required for SaaS applications.

b. Functional Requirements (FR)

Functional Requirements (FR) further refine the User Requirements. The Functional Requirements are key to define the agreed system operations, functions, security, processes, regulatory expectations and ensure that all user requirements are delivered and met. Data sensitivity contributes to the determination of the appropriate cybersecurity controls needed to protect the data. SaaS applications may leverage functional requirements from the software vendor.

c. Electronic Records/Electronic Signatures

For Quality applications ensure applicable e-Records and e-Signatures (ERES) requirements are included in the Requirement Specification.

Quality applications that create, modify, maintain, or transmit the following electronic quality records and electronic signatures under predicate rules must document and test applicable requirements for:

- Records that are in place of paper format
- Records, in addition to paper format (hybrid records), that are relied on to perform regulated activities
- Records submitted to regulatory agencies, under predicate rules (even if such records are not specifically identified in agency regulations)
- Signatures intended to be the equivalent of handwritten signatures, initials, and other general signings

d. Requirements Risk Rating/Test Assurance Determination

The Requirements Risk Rating / Test Assurance Level determination is required for computer applications with an SRC classification of Quality Critical that are defined as Configured or Custom in the SRC. Functional requirements are subjected to a risk rating analysis in the Requirements Specification to determine an appropriate Test Assurance level. Each requirement is evaluated for its failure's potential impact to compromise patient safety, degrade product quality, erode data integrity, security or miss a non-quality regulatory expectation per the following table.

This information is confidential to Abbott. The user is responsible for using the appropriate version of this document.

- Computer applications with an SRC of Quality Critical (COTS) or Quality Non-Critical (not custom) or Regulated, may elect to assess the Risk Rating and aligned Test Assurance level for assessed requirements. Alternatively, all requirements can be tested at the SRC classification instead of the requirement classification.
- When performing a Risk Rating analysis to determine the Test Assurance level, only perform against the functional requirements, as the user requirements "inherit" the highest collective rating of associated functional requirements.
- When functional requirements are not delivered for a computer system, the user requirements may be subjected to the same analysis (e.g. SaaS systems).

Requirement Classification	Impact Level	Criteria
Quality Critical (QC)	GxP Direct	• Direct impact on patient, personnel, public and/or product safety, product quality, efficacy and/or performance • Quality data that is analyzed, retrieved, manipulated, stored, reported and/or maintained that directly affect patient, personnel, public and/or product safety, product quality, efficacy, and/or performance. Consideration must be given to missing, incorrect or changed data where it may result in an inappropriate decision or action
Quality Non-Critical (QNC)	GxP Indirect	• Indirect impact on patient, personnel, public and/or product safety, product quality, efficacy and/or performance • Quality data that is analyzed, retrieved, manipulated, stored, reported and/or maintained that indirectly affect patient, personnel, public and/or product safety, product quality, efficacy, and/or performance. Consideration must be given to missing, incorrect or changed data where it may result in an inappropriate decision or action • Functionality that provides for compliance with governmental regulations not classified as "critical" above, and/or Functionality that controls records required to meet regulatory requirements
Non-Quality Highly Restricted (NQHR), Restricted (NQR)	Regulated (not GxP)	A requirement with no GxP impact but intended to satisfy a requirement from any other regulatory body or a computer function that accesses Restricted or Highly Restricted data.
Non-Quality Other Regulation (NQOR)	Regulated (not GxP	A requirement with no GxP and no Privacy regulatory impact but intended to satisfy a requirement from any other regulatory body or standard.
Non-Quality Highly Restricted (NQHR), Restricted (NQR)	Not Regulated Business Sensitive	Requirement that does not support any regulation, but supports data considered Business Sensitive.

This information is confidential to Abbott. The user is responsible for using the appropriate version of this document.

Requirement Classification	Impact Level	Criteria
Non-Regulated (NR)	Not Regulated and Not Business Sensitive	• No impact on manufacturing, labeling, product testing, on patient, personnel, public and/or product safety, product quality, efficacy and/or performance • Data that is analyzed, retrieved, manipulated, stored, reported and/or maintained is not related to patient, personnel, public and/or product safety, product quality, efficacy and/or performance • No associated regulatory requirements • No control of personal information falling into the “Privacy Highly Restricted” or “Privacy Restricted data sensitivity classifications • No control of business sensitive information falling into the “Highly Restricted” or “Restricted” data sensitivity classifications, and • No impact on external financial reporting systems

After determining the Requirement Classification / Impact Level, the requirement is evaluated to determine its Complexity (Implementation Method).

Complexity (Implementation Method)	Criteria
Custom	Functionality that is custom developed/coded/scripted for Abbott to meet Abbott’s business requirements.
Configured	Functionality that is modeled / configured to meet Abbott’s requirements.
COTS	Functionality that is commercial off the shelf/out of the box. No configurations or customizations are necessary for this functionality to be used by Abbott.

The minimum Test Assurance Level is obtained by examining the intersection of Requirement Classification / Impact Level and Complexity (Implementation Method), which also corresponds to the recommended testing (Scripted / Scenario Based / UAT). Requirements are assessed considering the most severe outcome resulting from the requirement’s potential to cause overall feature level failure. The table reflects minimum testing method; a more robust method can be applied. The amount of objective evidence should increase as the regulatory applicability and complexity increases.

This information is confidential to Abbott. The user is responsible for using the appropriate version of this document.

Test Assurance Risk Matrix

Requirement Classification	Impact Level	Complexity(Implementation Method)		
		COTS/OOTB	Configured	Custom
Quality Critical (QC)	GxP Direct	Scenario Based	Scripted	Scripted
Quality Non-Critical (QNC)	GxP Indirect	Scenario Based	Scenario Based	Scripted
Non-Quality Highly Restricted (NQHR) / Restricted (NQR) Other Regulated (NQOR)	Regulated (not GxP) Not Regulated (Business Sensitive)	Scenario Based	Scenario Based	Scenario Based
Non-Regulated (NR)	Not Regulated and Not Business Sensitive	UAT	UAT	Scenario Based

Objective Evidence

Scripted	Test Assurance is conducted by scripted testing. Scripted template is used to test functionality with step-by-step instructions (traditional validation). Testing includes; Expected Results, Actual Results and Pass/Fail for each test step with objective evidence at logical steps.
Scenario Based	Scenario Based is used to test functionality with less prescriptive (not step-by-step) instructions. The tester's actions are not prescribed by detailed instructions in advance of test execution to the same extent as scripted testing. Testing includes Pass / Fail for each test step with objective evidence at key points or if failures are encountered in testing.
UAT	Test Assurance is supported by User Acceptance Testing (UAT), executed using Scenario Based Testing.

4.2.2.2 Functional Risk Assessment (FRA)

A Functional Risk Assessment (FRA) is created and maintained for quality critical systems. Based on the URS and/or FRS and the related business process, it can be determined which areas or individual functionalities have a critical quality risk and require mitigation strategies to be implemented to reduce the risk in these areas and for the entire system. The Functional Risk Assessment should be delivered and reviewed prior to the design phase of the project to ensure risk mitigations that require testing can be incorporated into the project's testing strategy.

4.2.2.3 Data Migration Plan

Procedure BTSQC09.05.023

The Data Migration Plan describes how to scope impacted data, data cleansing, use and qualification needs of archival and migration tools, data cleansing, data mapping, data transformations, transfer and testing activities to be executed in the QA versus Production environment, assessing data criticality and associated risk, and determining associated data sampling / verification sizes.

Data sampling can be executed per ISO 2859.1, or equivalent, with Low Risk mapped to Special Inspection Level 1, Medium Risk mapped to General Inspection Level 1 and High Risk mapped to General Inspection Level 2.

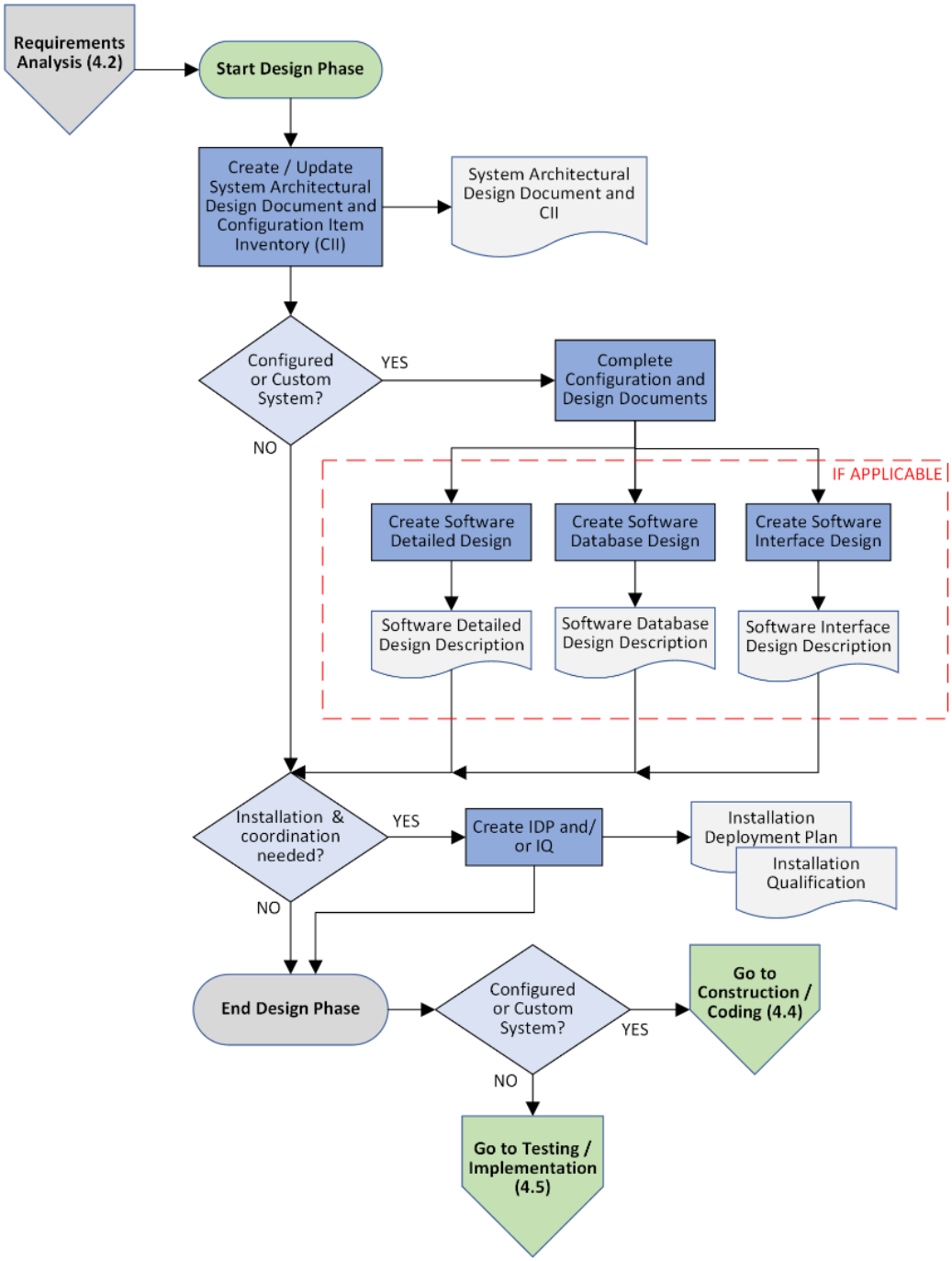
For privacy-impacted applications, the data archival and / or migration must ensure the privacy data is adequately protected at all stages of the migration including any requirements relating to the location of the data and who may access the data. Consult data privacy and security SMEs.

This information is confidential to Abbott. The user is responsible for using the appropriate version of this document.

4.3 Design Phase

This phase translates the functional, regulatory and security control requirements from approved Requirements Specification into a software design solution intended to automate a business process and/or transfer data to another system.

4.3.1 Design Process Flow



This information is confidential to Abbott. The user is responsible for using the appropriate version of this document.

4.3.2 Deliverables

4.3.2.1 Configuration and Design Documents

Configuration and design documents are intended to represent a technical "conversation" between delivery and support personnel and to facilitate an understanding of how functions are designed and built. These documents need to provide sufficient detail to support the computer system, make changes and investigate and resolve issues.

a. Architectural Design Document and CII

The Architectural Design Document and CII describes the application's architecture from a system, software, application, data, and interface perspective. It also defines high-level design and / or configuration parameters, including cybersecurity (when required for Restricted or Highly Restricted data), security, performance, and resiliency (i.e., availability, backups, and recovery), intended to satisfy the system's functional behaviors in support of business needs. The Configuration Item Inventory (CII) section is required for Quality systems and is used to identify and document configuration items for the system.

b. Software Database Design Description

The Software Database Design Description is used to detail all aspects related to the computer system's database, including entity, attribute, table, table relationships, columns, records, views triggers, indexes, sequences, database security and database sizing.

c. Software Detailed Design Description

The Software Detailed Design Description defines the modules or components and the functions of a configurable or custom system. It also defines detailed design and / or configuration parameters, including functional security, performance, and resiliency (i.e., availability, backups, and recovery).

d. Software Interface Design Description

The Software Interface Design Description defines interface(s) that send / receive information to / from a configurable or custom system and other systems, including both internal (inside the system) and external (outside the system) interfaces, and those where the application is acting as a service provider (push) or as a consumer of a service (pull). It also defines user interfaces to and from the system. Consider cybersecurity requirements during design and development of interfaces as applicable.

4.3.2.2 Installation and Deployment Plan (IDP)

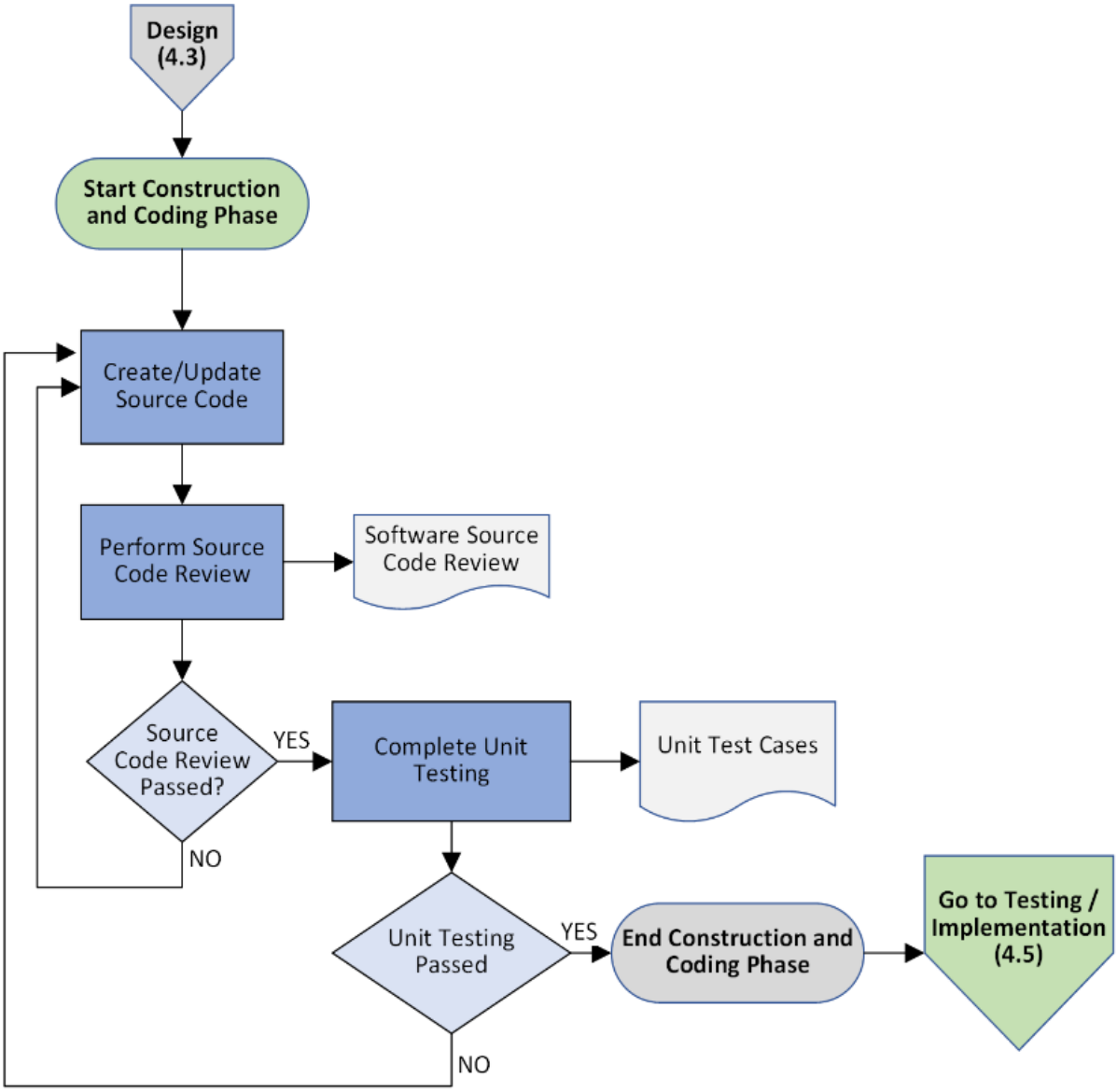
The Installation and Deployment Plan may be used when coordination of installation activities across functions or sequences is warranted. The IDP describes the process, sequence, timing, and strategy to deploy a new system or maintenance release to the production environment. This document defines deliverables (e.g., IQs) to be presented to the organization responsible for software installation, intended to install and verify the application. In addition, supporting user documentation, data migration activities as defined in the plan, and security requirements should be included as applicable.

This information is confidential to Abbott. The user is responsible for using the appropriate version of this document.

4.4 Construction and Coding Phase

During this phase, developers build configurations / customizations based on the agreed-upon requirements and / or configuration / design specifications. Developers may coordinate code reviews and perform developer testing to ensure the system meets its intended use. If unit / developer testing is performed, it is recommended to execute as scenario-based testing.

4.4.1 Construction and Coding Process Flow



This information is confidential to Abbott. The user is responsible for using the appropriate version of this document.

4.4.2 Deliverables

4.4.2.1 Source Code Management / Software Source Code Review

Procedure BTSQC09.05.022

Source code developed for custom functionality should be developed to intended purposes, follow expected Abbott and system-specific standards, and be deployed in a consistent and controlled manner. For custom developed code, a peer Software Source Code Review should be performed prior to unit testing, qualification, or validation activities. Code Reviews are required for quality-critical custom code and recommended for all custom code and complex configured functions. An automated code management tool may be leveraged in lieu of a manual Code Review.

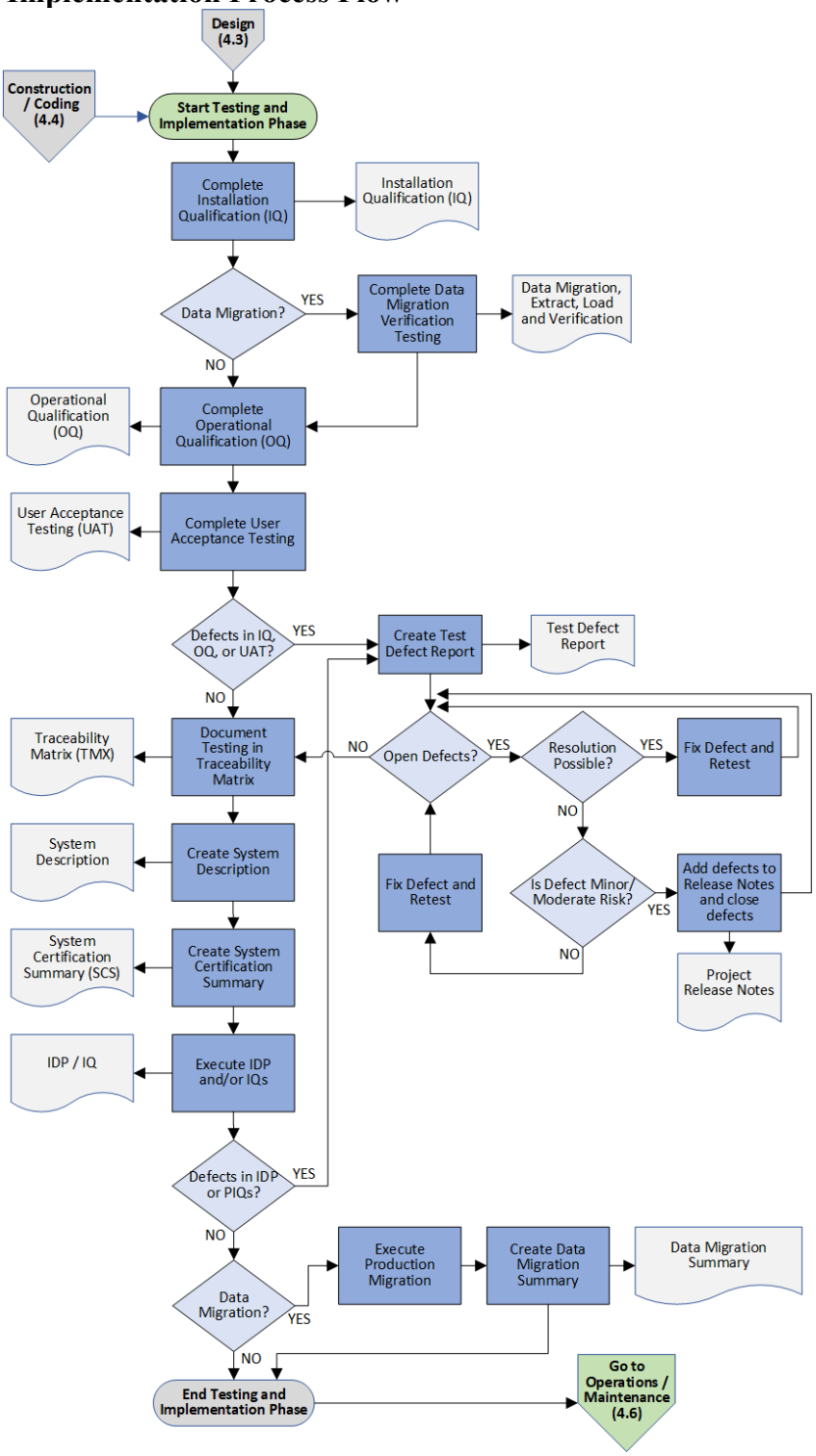
4.4.2.2 Design Verification / Unit Testing

Design verification / unit testing is a type of software testing where individual units or components of a software are tested to confirm that each unit of the software code performs as expected. Software Unit Tests are used to test the design accuracy of modules in custom developed software, and / or complex configured functionality. Unit testing is required for quality-critical custom functions. A unit may be an individual function, method, procedure, module, or object.

This information is confidential to Abbott. The user is responsible for using the appropriate version of this document.

4.5 Testing and Implementation Phase

4.5.1 Testing and Implementation Process Flow



This information is confidential to Abbott. The user is responsible for using the appropriate version of this document.

4.5.2 Deliverables

4.5.2.1 Installation Qualification

An Installation Qualification is used to document the installation and verification of the application software in a controlled environment.

4.5.2.2 Data Migration Extract, Load and Verification

This test case is used to extract, load, and verify computer system data in a controlled manner. The details of testing and verification are detailed in the associated Data Migration Plan.

4.5.2.3 Operational Qualification (OQ) (e.g., Test Protocol)

Procedure BTSQC09.05.002

The Operational Qualification (Test Protocol) is used to document the validation or verification testing of the application software. Validation / verification testing is designed and executed to provide documented evidence that the software and / or computerized application will consistently achieve its predetermined specified requirements and supports the documented intended use.

a. Scripted Testing

Scripted testing includes detailed step instructions and defines the Expected Results.

b. Scenario Based Testing

Scenario Based testing is dynamic testing in which the tester's actions are not prescribed by detailed, step-level instructions.

c. Technical Security Controls Testing

Technical security controls testing confirms DPP controls identified for the application have been applied.

4.5.2.4 User Acceptance Testing (UAT)

User Acceptance Testing confirms that the system meets the business intended use and user needs e.g., executing business scenarios within a controlled environment.

4.5.2.5 Test Defect Report

Procedure BTSQC09.05.025

A Test Defect Report is used to identify, track, and resolve unexpected system events encountered during the Testing and Implementation Phase in a controlled, non-production environment. Test Defect Reports are often triggered by the actual result of testing not matching the expected result or by observed system behavior not satisfying the test objective. All Test Defect Reports must be resolved and closed or dispositioned to an alternate record and closed prior to releasing the computer application to end users.

4.5.2.6 Project Release Notes

The Project Release Notes is used to record computer system test defects and features, as defined in the Requirements Specification (either URS or FRS), not delivered at time of system certification as well as the business workaround and acceptance of business impact.

This information is confidential to Abbott. The user is responsible for using the appropriate version of this document.

4.5.2.7 Traceability Matrix (TM)

The Traceability Matrix is a document that links requirements throughout the system lifecycle process and ensure that all requirements defined for a regulated system are tested in the test protocols or fulfilled by an external process (e.g., via procedure). Initiate the Traceability Matrix during the Requirements Phase with the URS to FRS traceability (if the URS and FRS are delivered as independent documents). The user and functional requirements are traced by design when using the Requirements Specification (RS) template. Continue updating the document throughout the SLC as indicated in the table below.

Note: Traceability may be delivered via an approved document or may be provided via alternate means (e.g., a naming convention, a validated tool with electronic linking of entities, etc.). The traceability method employed by the project must be identified in the approved plan document or a controlled procedure (SOP).

Requirement Classification	Delivered through:	URS to FRS*	FRS toOQ	FRS to Design	Design toUnit test
Quality Critical	Custom Code	Required	Required	Required	Required
	Configured	Required	Required	Required	Optional
	COTS	Required if Applicable	Required**	Not Applicable	Not Applicable
Quality Non-Critical	Custom Code	Required	Required	Optional	Optional
	Configured	Required	Required	Optional	Optional
	COTS	Required if Applicable	Required**	Not Applicable	Not Applicable
Non-Quality Highly Restricted/ Restricted/ Other Regulated	Custom Code	Required	Required	Optional	Optional
	Configured	Required	Required	Optional	Optional
	COTS	Required if Applicable	Required**	Not Applicable	Not Applicable
Non-Regulated	Custom Code	Optional	Required	Optional	Optional
	Configured	Optional	Required	Optional	Optional
	COTS	Optional	Required**	Not Applicable	Not Applicable

* If URS and FRS are independent deliverables.
** May be traced as URS to UAT

This information is confidential to Abbott. The user is responsible for using the appropriate version of this document.

4.5.2.8 System Description (SD)

A System Description must be created for all Quality applications and is intended to facilitate continuous monitoring and audit readiness. The SD document is a concise overview of the computer system with details about the system, supported quality processes, software validation and procedural references. The SD is updated for change requests and releases as defined in application documentation.

4.5.2.9 System Certification Summary

The System Certification Summary (SCS) recaps the documents and activities completed for the project and is required for Regulated (including Quality) applications to ensure that software and software-related services meet the requirements of the accepting division and / or site and satisfies the regulatory requirements, as applicable. The SLC package is reviewed and summarized. All test defects encountered in controlled environments during testing, and deviations from the Validation Plan or Verification Plan must be documented.
Upon approval of the SCS, the application change is released to the production environment. After production build and verification, it is recommended to revise the SCS to include details of production installation, verification, and defects, if any, to close the computer system validation and project. The last step is to close the project validation package.

Conditional certification must be approved by the Validation Review Board (VRB). Conditional certifications must identify conditions of certification, justification, and timing. If the targeted completion dates need to be extended, the certification summary may be updated with the new target completion dates and approved. When conditions are complete, the certification summary is updated to document conditional items are closed and approved.

4.5.2.10 Data Migration Summary

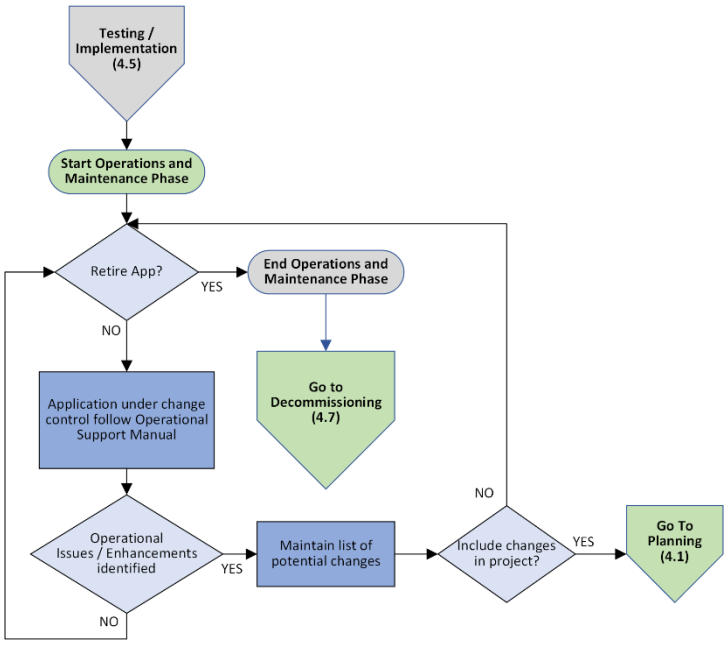
Procedure BTSQC09.05.023
The Data Migration Summary describes and summarizes the results of the activities as documented within the Data Migration Plan, including any deviations from the plan.
Approval of the Data Migration Summary signifies that work has been successfully completed and the data in scope has been approved for use in production.
When the change occurring is only a migration with no new or revised functionality, this document can replace the System Certification Summary.

This information is confidential to Abbott. The user is responsible for using the appropriate version of this document.

4.6 Operations and Maintenance Phase

The Operations and Maintenance Phase begins after the system is implemented in the production environment.

4.6.1 Operational and Maintenance Process Flow



4.6.2 Deliverables

4.6.2.1 Change Control

Procedure BTSQC09.05.008 Changes to applications include standard changes, maintenance changes and emergency changes. Change control in the maintenance phase can include a Project Change Request or Maintenance Change Request. These Change Control documents evaluate the impact of changes on applications. Impact of change should include review of interfaces, architecture, infrastructure, documentation, data migration, system risk classification, DLM, BIA, quality risks and regulatory risks.

4.6.2.2 Maintenance Plan

Procedure BTSQC09.05.008

A Maintenance Plan can be used to define a tailored approach for supporting the software / system in the maintenance phase of system life cycle (SLC) after a release is in production. The Maintenance Plan should be viewed as an extension of the Validation Plan and is approved by the same roles as the Validation Plan.

In lieu of a dedicated Maintenance Plan, applications may follow a Maintenance Procedure (i.e., BTSQC09.05.008, BTS SLC Maintenance Change Procedure, or equivalent)

This information is confidential to Abbott. The user is responsible for using the appropriate version of this document.

4.6.2.3 Continuous Monitoring of the Validated State of Computerized Applications (Periodic Review)

BTSQC09.05.GL006

Continuous monitoring of applications confirms a BTS supported quality system has been maintained in a validated state where required by regulation. This review is conducted on a regular basis under Management Review, Change Control and other controls.

4.6.2.4 Data Changes

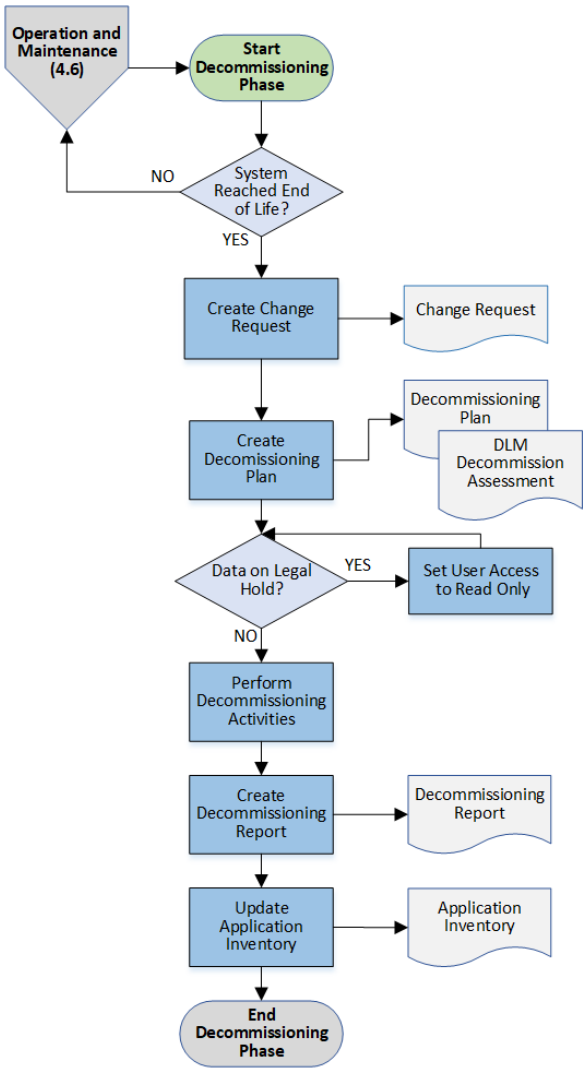
Procedure BTSQC09.05.060

Production data changes for regulated and/or quality non-product applications follow BTSQC09.05.060. Examples of data changes can include: removing duplicate records from a training management system, changing data in a database, updating the state of records in database.

This information is confidential to Abbott. The user is responsible for using the appropriate version of this document.

4.7 Decommissioning Phase

4.7.1 Decommissioning Process Flow



This information is confidential to Abbott. The user is responsible for using the appropriate version of this document.

4.7.2 Deliverables

4.7.2.1 Change Request

When decommissioning an application, a change request is used to complete all activities to remove the application from production and to close any project documentation.

4.7.2.2 Decommissioning Plan

Decommissioning Plan is required to document activities to be performed to remove an application from service. Decommission Plan should include but is not limited to the following steps: Obsolete the SRC; close all Change Requests, update application inventory to decommissioned status and remove from Validation Master Plan (VMP). Before a Decommissioning Plan is approved, a DLM Decommission Assessment must be performed to identify retention requirements and legal hold order applicability. Following the DLM assessment policy I1-04, source data must be preserved over the retention period.

4.7.2.3 Data Lifecycle Management (DLM) Decommission

Policy I1-04

A Data Lifecycle Management (DLM) Decommission Assessment provides the requirements for managing the retention of electronic records. Regulated (quality and non-quality) applications set to become inactive must be assessed for record retention requirements.

4.7.2.4 Decommissioning Report

The Decommissioning Report documents the results of the system decommissioning activities as per the approved plan. Any deviations must be documented, justified, and approved. Each activity identified in the plan must be addressed.

Note: After the report is approved, the BTS System of Record and application inventory must be updated to reflect the decommissioned status.

5.0 Appendices

Appendix A	Author and Approval Matrix for Quality Applications
Appendix B	Author and Approval Matrix for Regulated (Non-Quality) Applications
Appendix C	Supplemental Policies/Process

This information is confidential to Abbott. The user is responsible for using the appropriate version of this document.

Appendix A: Author and Approval Matrix for Quality Applications

SLC Deliverable	Template/ Procedure Reference	Suggested Author	Minimum Approvals				COTS	Config/ Custom
			Process Owner	Compliance	System Owner	Technical Resp[9]		
Planning Phase								
System Risk Classification	BTSQC09.05-A	Primary IT Lead	BPO	SQA Lead SME [1]	IT Support Owner	N/A	R	R
BTS Computerized Tool Validation	BTSQC09.05.024	Primary IT Lead	Per BTS Computerized Tool Validation Procedure				RIA	RIA
Change Request	BTSQC09.05-B	Validation Lead	BPO	SQA Lead [2,7]	System Owner[6]	N/A	R	R
Validation Plan	BTSQC09.05-D	Validation Lead	BPO	SQA Lead [2]	System Owner[6]	N/A	RIA	RIA
Data Protection Plan	BTSQC09.05-G	Primary IT Lead	BPO	Privacy SME	N/A	Architect	RIA	RIA
Requirements Analysis Phase								
Requirements Specification	BTSQC09.05-AT	Primary IT Lead	BPO	SQA	System Owner[6]	N/A	R	R
Functional Risk Assessment	BTSQC09.05-AE	Validation Lead	BPO	SQA	Primary IT Lead	N/A	RIA	RIA
Data Migration Plan	BTSQC09.05-L	Validation Lead	BPO	SQA	System Owner[6]	Technical	RIA	RIA
Design Phase								
Architectural Design Document /CII	BTSQC09.05-AS	IT Technical Team	N/A	SQA	N/A	Architect	R[4]	R
Software Database Design Description	BTSQC09.05-N	IT Technical Team	N/A	N/A	N/A	Architect	N/A	RIA
Software Detailed Design Description	BTSQC09.05-O	IT Technical Team	N/A	N/A	N/A	Architect	N/A	RIA
Software Interface Design Description	BTSQC09.05-P	IT Technical Team	N/A	N/A	N/A	Architect	RIA	RIA

This information is confidential to Abbott. The user is responsible for using the appropriate version of this document.

Appendix A: Author and Approval Matrix for Quality Applications

SLC Deliverable	Template/ Procedure Reference	Suggested Author	Minimum Approvals				COTS	Config/ Custom
			Process Owner	Compliance	System Owner	Technical Resp[9]		
Installation Deployment Plan (IDP) (Pre and Post)	BTSQC09.05-K	Primary IT Lead	BPO	SQA[8]	System Owner[6]	N/A	RIA	RIA
Construction and Coding Phase								
Software Source Code Review	BTSQC09.05-R	IT Technical Team	N/A	N/A	N/A	Architect	N/A	R[5]
Software Unit Test	BTSQC09.05-T	IT Technical Team	N/A	N/A	N/A	Architect	N/A	R[5]
Testing and Implementation Phase								
Installation Qualification (IQ) (Pre and Post)	BTSQC09.05-X	IT Delivery Lead	N/A	SQA[8]	N/A	Architect	RIA	RIA
Data Migration, Extract and Load Verification (Pre and Post)	BTSQC09.05-AQ	Primary IT Lead	BPO	SQA	System Owner [6]	Technical	RIA	RIA
Scripted Test Case (Pre and Post)	BTSQC09.05-Y	Validation Lead	N/A	SQA[3]	N/A	Technical[12]	RIA	R
Scenario Based Test Case (Pre and Post)	BTSQC09.05-AU	Validation Lead	N/A	SQA[3]	N/A	Technical[12]	R	R
Technical Control Testing (Pre and Post)	BTSQC09.05-AAI	Architect	N/A	N/A	N/A	Technical	RIA	RIA
User Acceptance Test (UAT) (Pre and Post)	BTSQC09.05-AU	Business Delegate	BPO	SQA[3]	N/A	N/A	R	R
Test Defect Report (Pre and Post)	BTSQC09.05-V	Validation Lead	N/A	SQA[3]	N/A	Technical[12]	RIA	RIA
Project Release Notes	BTSQC09.05-W	Validation Lead	BPO	SQA	System Owner[6]	N/A	RIA	RIA
Traceability Matrix	BTSQC09.05-U	Validation Lead	N/A	SQA	N/A	Architect	R	R
System Description	BTSQC09.05-AAH	Validation Lead	BPO	SQA	System Owner[6]	N/A	R	R

This information is confidential to Abbott. The user is responsible for using the appropriate version of this document.

Appendix A: Author and Approval Matrix for Quality Applications

SLC Deliverable	Template/ Procedure Reference	Suggested Author	Minimum Approvals				COTS	Config/ Custom
			Process Owner	Compliance	System Owner	Technical Resp[9]		
System Certification Summary (SCS)	BTSQC09.05-AA	SQA[11]	BPO	SQA Lead[2]	System Owner[6]	N/A	R	R
Data Migration Summary	BTSQC09.05-AI	Validation Lead	BPO	SQA	System Owner[6]	Technical	RIA	RIA
Operations and Maintenance Phase								
Maintenance Plan[10]	BTSQC09.05-AH	IT Delivery Lead	BPO	SQA	System Owner[6]	N/A	O	O
Decommissioning Phase								
Decommissioning Plan	BTSQC09.05-AJ	Validation Lead	BPO	SQA	IT Support Owner	Technical	RIA	RIA
Decommissioning Report	BTSQC09.05-AK	Validation Lead	BPO	SQA	IT Support Owner	Technical	RIA	RIA

Appendix A: Author and Approval Matrix for Quality Applications

LEGEND

R	Required	RIA	Required If Applicable	O	Optional	N/A	Not Applicable
BPO	Business Process Owner	SQA	Software Quality Assurance	SME	Subject Matter Expert		

Note [1] = Regulatory SME (e.g., Privacy, SOX, PCI) approval is required for regulatory domains in scope of the application

Note [2] = SQA Lead required for Quality Critical applications; SQA can approve quality non-critical applications

Note [3] = SQA pre and post approval required for regulated functions. No compliance approval needed for non-regulated functions

Note [4] = Configuration Inventory Item (CII) section not required for COTS applications with no configuration

Note [5] = Required for quality critical custom functions. Recommended for complex configured and all custom functions

Note [6] = System Owner may be the IT Delivery Owner or the IT Support Owner depending on project delivery versus operational support and if an operational release support requires a delivery team.

Note [7] = SQA Lead required for Change Request that supports an Emergency Change

Note [8] = SQA approves the IQ and/or IDP as defined in the plan document

Note [9] = Technical Responsible role can be fulfilled by Architect or Technical

Note [10] = Maintenance Plan SOP (BTSQC09.05.008) can be used or specific application maintenance plan (BTSQC09.05-AH)

Note [11] = IT can author System Certification Summary

Note [12] = System Owner or Technical can approve testing and testing defects

This information is confidential to Abbott. The user is responsible for using the appropriate version of this document.

Appendix B: Author and Approval Matrix for Regulated Non-Quality Applications

SLC Deliverable	Template/ Procedure Reference	Suggested Author	Minimum Approvals				COTS	Config/ Custom
			Process Owner	Compliance	System Owner	Technical Resp[6]		
Planning Phase								
System Risk Classification	BTSQC09.05-A	Primary IT Lead	BPO	SQA Lead SME [1]	IT Support Owner	N/A	R	R
BTS Computerized Tool Validation	BTSQC09.05.024	Primary IT Lead	Per BTS Computerized Tool Validation Procedure				RIA	RIA
Change Request	BTSQC09.05-B	Primary IT Lead	BPO	CA[7]	System Owner[3]	N/A	R	R
Verification Plan	BTSQC09.05-C	Primary IT Lead	BPO	CA	System Owner[3]	N/A	RIA	RIA
Data Protection Plan	BTSQC09.05-G	Primary IT Lead	BPO	Privacy SME	N/A	Architect	RIA	RIA
Requirements Analysis Phase								
Requirements Specification	BTSQC09.05-AT	Primary IT Lead	BPO	CA	System Owner [3]	N/A	R	R
Data Migration Plan	BTSQC09.05-L	Primary IT Lead	BPO	CA	System Owner[3]	Technical	RIA	RIA
Design Phase								
Architectural Design Document/CII	BTSQC09.05-AS	IT Technical Team	N/A	CA	N/A	Architect	RIA[4]	R[4]
Software Database Design Description	BTSQC09.05-N	IT Technical Team	N/A	N/A	N/A	Architect	N/A	RIA[6]
Software Detailed Design Description	BTSQC09.05-O	IT Technical Team	N/A	N/A	N/A	Architect	N/A	RIA[6]
Software Interface Design Description	BTSQC09.05-P	IT Technical Team	N/A	N/A	N/A	Architect	N/A	RIA[6]
Installation Deployment Plan (IDP) (Pre and Post)	BTSQC09.05-K	Primary IT Lead	BPO	CA[8]	System Owner [3]	N/A	RIA	RIA

This information is confidential to Abbott. The user is responsible for using the appropriate version of this document.

Appendix B: Author and Approval Matrix for Regulated Non-Quality Applications

SLC Deliverable	Template/ Procedure Reference	Suggested Author	Minimum Approvals				COTS	Config/ Custom
			Process Owner	Compliance	System Owner	Technical Resp[6]		
Construction and Coding Phase								
Software Source Code Review	BTSQC09.05-R	IT Technical Team	N/A	N/A	N/A	Architect	N/A	O [5]
Unit Test	BTSQC09.05-T	IT Technical Team	N/A	N/A	N/A	Architect	N/A	O [5]
Testing and Implementation Phase								
Installation Qualification (Pre and Post)	BTSQC09.05-X	IT Delivery Lead	N/A	CA[8]	N/A	Architect	RIA	RIA
Data Migration, Extract and Load Verification	BTSQC09.05-AQ	Primary IT Lead	BPO	CA	System Owner [3]	Technical	RIA	RIA
Technical Control Testing Template (Pre and Post)	BTSQC09.05-AAI	Architect	N/A	N/A	N/A	Technical	RIA	RIA
Scenario Based Test Case (Pre and Post)	BTSQC09.05-AU	Validation Lead	N/A	CA[2]	N/A	Technical[10]	RIA	R
User Acceptance Test (UAT) (Pre and Post)	BTSQC09.05-AU	Business Delegate	BPO	CA[2]	N/A	N/A	R	R
Test Defect Report (Pre and Post)	BTSQC09.05-V	Validation Lead	N/A	CA[2]	N/A	Technical[10]	RIA	RIA
Project Release Notes	BTSQC09.05-W	Primary IT Lead	BPO	CA	System Owner [3]	N/A	RIA	RIA
Traceability Matrix	BTSQC09.05-U	Primary IT Lead	N/A	CA	N/A	Architect	R	R
System Certification Summary (SCS)	BTSQC09.05-AA	Primary IT Lead	BPO	CA	System Owner[3]	N/A	R	R
Data Migration Summary	BTSQC09.05-AI	Primary IT Lead	BPO	CA	System Owner[3]	Technical	RIA	RIA

This information is confidential to Abbott. The user is responsible for using the appropriate version of this document.

Appendix B: Author and Approval Matrix for Regulated Non-Quality Applications

SLC Deliverable	Template/ Procedure Reference	Suggested Author	Minimum Approvals				COTS	Config/ Custom
			Process Owner	Compliance	System Owner	Technical Resp[6]		
Operations and Maintenance Phase								
Maintenance Plan[9]	BTSQC09.05-AH	Primary IT Lead	BPO	CA	System Owner [3]	N/A	O	O
Decommissioning Phase								
Decommissioning Plan	BTSQC09.05-AJ	Primary IT Lead	BPO	CA	IT Support Owner	N/A	RIA	RIA
Decommissioning Report	BTSQC09.05-AK	Primary IT Lead	BPO	CA	IT Support Owner	N/A	RIA	RIA

Appendix B: Author and Approval Matrix for Regulated Non-Quality Applications

LEGEND

R	Required	RIA	Required If Applicable	O	Optional	N/A	Not Applicable
BPO	Business Process Owner	CA	Compliance Analyst	SME	Subject Matter Expert		

Note [1] = Regulatory SME (e.g., Privacy, SOX, PCI) approval is required for regulatory domains in scope of the application

Note [2] = CA pre and post approval required for regulated functions. No compliance approval needed for non-regulated functions.

Note [3] = System Owner may be the IT Delivery Owner or the IT Support Owner depending on project delivery versus operational support and if an operational release support requires a delivery team

Note [4] = Configuration Inventory Item (CII) section not required

Note [5] = Recommended for complex configured and all custom functions

Note [6] = Technical Responsible can be Architect or Technical

Note [7] = SQA Lead required for Change Request that supports an Emergency Change

Note [8] = CA approves the IQ and/or IDP as defined in the plan document

Note [9] = Maintenance Plan SOP (BTSQC09.05.008) can be used or specific application maintenance plan (BTSQC09.05-AH)

Note [10]= System Owner or Technical can approve testing and testing defects

Non-Regulated applications require a System Risk Classification, DLM Assessment, BIA, and Data Protection Plan as applicable

This information is confidential to Abbott. The user is responsible for using the appropriate version of this document.

Appendix C: Supplemental Policies/Process (not controlled by BTS SLC)

Supplemental Process	Policy/Process Reference
Technology Vendor Risk Management Process	BTSQC11.01
BTS Supplier Quality Assurance Process	BTSQC11.02
Cybersecurity Vendor Risk Management Process	BTSQC19.21
Data Lifecycle Management (DLM) Policy	I1-04
Information Classification Policy	I1-05
IT Recovery Plan (includes Business Impact Analysis (BIA)/ Disaster Recovery Plan)	4-070
Digital Marketing	MKT03
Privacy Risk Assessment(PRA)	GLB-PRIVACY-RA-PROC

END OF DOCUMENT

This information is confidential to Abbott. The user is responsible for using the appropriate version of this document.

Document ID: BTSQC09.05
Control Site: BTS - IT (QA)

Info	
Document ID:	BTSQC09.05
Controlled Document Title:	BTSQC09.05
Title:	BTS IT Computerized System Life Cycle
Document Category:	Quality Systems Documents
Document Type:	Process
Document Type Details:	
Major Version:	13
State:	Effective
Affecting Change Element:	New Version
Approved Date:	31-Oct-2024
Effective Date:	01-Dec-2024
Effective Date Type:	Effective Date
Division:	Business and Technology Services
Control Site:	BTS - IT (QA)
Use Overlay:	General - Effective Date
Translation Source:	
Language:	English(en)
Authors:	
Owner Functional Area:	Enterprise Strategy, Planning, Quality and Compliance (ESP, Q&C)
Departments/Areas Affected:	
Security:	No / Public
Record Series Code:	
Retention Start Date:	
Retention End Date:	
No Purge:	No
Legacy Object ID:	
Legacy Information:	
Periodic Review	
Disable Periodic Review:	No
Last Review Date:	04-Oct-2022
Review Interval:	36
Next Review Date:	04-Oct-2025
Subject Matter Expert:	Lagrou, Linda E
Post Approval	
Business Type:	
Governing Document:	AQ09.A, AQ0905.A
Reference Document:	AQ090501.A, BTSQC09.05, BTSQC09.05-A, BTSQC09.05-B, BTSQC09.05-C, BTSQC09.05-D, BTSQC09.05-G, BTSQC09.05-K, BTSQC09.05-L, BTSQC09.05-N, BTSQC09.05-O, BTSQC09.05-P, BTSQC09.05-R, BTSQC09.05-T, BTSQC09.05-U, BTSQC09.05-V, BTSQC09.05-W, BTSQC09.05-X, BTSQC09.05-Y, BTSQC09.05-AA, BTSQC09.05-AE, BTSQC09.05-AH, BTSQC09.05-AI, BTSQC09.05-AJ, BTSQC09.05-AK, BTSQC09.05-AQ, BTSQC09.05-AS, BTSQC09.05-AT, BTSQC09.05-AU,

Source Document:	BTSQC09.05-AAH, BTSQC09.05-AAI, BTSQC09.05.002, BTSQC09.05.007, BTSQC09.05.008, BTSQC09.05.022, BTSQC09.05.023, BTSQC09.05.024, BTSQC09.05.025, BTSQC09.05.060, BTSQC09.05.GL006, BTSQC11.01, BTSQC11.02, BTSQC19.21, GLB-PRIVACY-RA-PROC, I1-02, I1-04, I1-05, 4-070, ISO2859.1, MKT03, QC10.11, Sarbanes Oxley (SOX)
Supplier Applicable Documents:	

END OF DOCUMENT

Abbott - Business and Technology Services
Change Request Properties

CR Number: CR-BTSQA-23-00000618
Control Site: BTS - IT (QA)

Initiator	
CR Initiator:	Veeravalli, Ramya
Initiator Phone Number:	N/A
Initiator Location:	J46
Initiator Division:	Business and Technology Services

CR Info	
CR Number:	CR-BTSQA-23-00000618
CR Submitted Date:	29-Oct-2024
CR Control Site:	BTS - IT (QA)
CR Division:	Business and Technology Services
CR Status:	Closed
CR Status Date:	31-Oct-2024
CR Approved Date:	31-Oct-2024

Change Info	
Security:	No / Public
Document ID / Title:	BTSQC09.05 / BTS IT Computerized System Life Cycle
Description of Change:	List of description of changes and justification of changes for BTSQC09.05 v13 Please see attached word document for more description of changes Fixed spelling and grammar through out the document. Made format changes in the document 1.3.1 In-Scope •Removed ‘Applications and computer’. •Removed “if applicable” statement from data protection plan. •Removed Privacy Risk .Assessment(PRA) . 1.3.2 Out of Scope •Removed product software. •Removed Data Change . 1.4 Responsibilities 1.Added statement under responsibilities and removed the same from the 2nd paragraph. 2.BPO- Removed a statement and added policy numbers. 3.SQA and SQA Lead- Updated role and content. 4.IT Support owner/Primary IT Lead – removed network security policy and replaced it with Abbott’s security requirements. 5.Val lead- ‘Lead’ is added at the end of the SQA role. 6.Architect/Technical- Removed ‘Architect’ from the role and updated content. 7.VRB –Lead is replaced with Manager. 3.1.6 Software as a Service Added clarification that for application classified as quality, vendor responsibilities must be defined.
Reason / Justification for Change:	Please see the attached word document in the table for more reason and justification for changes. Section 1.3.1 - •Redundant •Privacy risk assessment is governed outside SLC and is not used in SLC execution. •Redundant Section 1.3.2 - •PSLC is obsolete.

•Data Change is now under BTSQC09.05 (new child SOP that will be included in metadata as reference doc).
1.What is the “driver” for the reason this document is being updated or new/needed. (CAPA, Project XXX)
(Why is the document needed, why is it important, what would happen if it did not exist?)-
CAPA = QR935377, QR935591
Process improvements
It is important to address non conformant issues per our processes. If the document didn’t exist we would not be compliant with AQR policies.
2.Impact to other areas including other divisions? (Yes or No) Explain.
Yes - The impact is limited to systems / areas in scope of BTS SLC (BTSQC09.05) – Software delivered/managed/maintained by BTS resources.
3.Impact to systems or applications? (Yes or No) Explain.
No - “No impact to any systems. Changes align with existing system configurations.”.
4.Impact to in process activities (e.g., SLC refresh, splitting of TVRM to include CVRM, etc.) Provide Implementation Strategy statement.
No, there are no current process initiatives impacted by these changes. The implementation strategy is addressed in the body of the document.
5.Impact to training, see next bullet.
No change to training the audience.

Was Impact Assessment Performed?: Yes

Training Requirements: Content Change

Comments: Linda Lagrou approving SME and Process Owner

No change to training the auidence

Effectivity

Effective Date Type: Effective Date

Implementation Type: Specify Implementation Date

Implementation Period:

Implementation Date: 01-Dec-2024

Implementation Justification: To meet CAPA Deadline

Change Documents

Document ID	Title	Document Type
BTSQC09.05	BTS IT Computerized System Life Cycle	Process

Supporting Documents

Name
BTSQC09.05 v13 Description of changes and justification 2024 (002)
BTSQC09.05 V13 Draft - IA
BTSQC09.05 v13 PRB Worksheet
BTSQC09.05 v13_Clean copy
BTSQC09.05_redline

Approvals

User Name	Name / Domain	Meaning of Signature	Functional Area	Date / Time
Lagrou, Linda E	ONEABBOTT\LAGR	Approver	SME	2024-10-31 20:01:14

CR Number: CR-BTSQA-23-00000618
Control Site: BTS - IT (QA)

	OLE			(UTC+00:00)
Rivi, Shari	ONEABBOTT\RIVISX	Approver	Quality	2024-10-31 21:45:35 (UTC+00:00)
Warnock, Elizabeth K	ONEABBOTT\WARNOEK	Approver	Quality Director	2024-10-31 22:05:49 (UTC+00:00)

External Approvals

< No External Approval Data to Display >

Linked CRs

< No Linked CR Data to Display >

END OF DOCUMENT